SR675 V3 system components

Components overview

Drive cages

The SR675 V3 supports the following drive cages:

- 4-DW GPU models: Eight 2.5-inch drive cage
- 8-DW GPU models: EDSFF E1.S drive cage or E3.S 1T drive cage
- SXM5 GPU models: 2.5-inch drive cage or E3.S 1T drive cage

Click the links to see schematic diagrams of the drive cages.

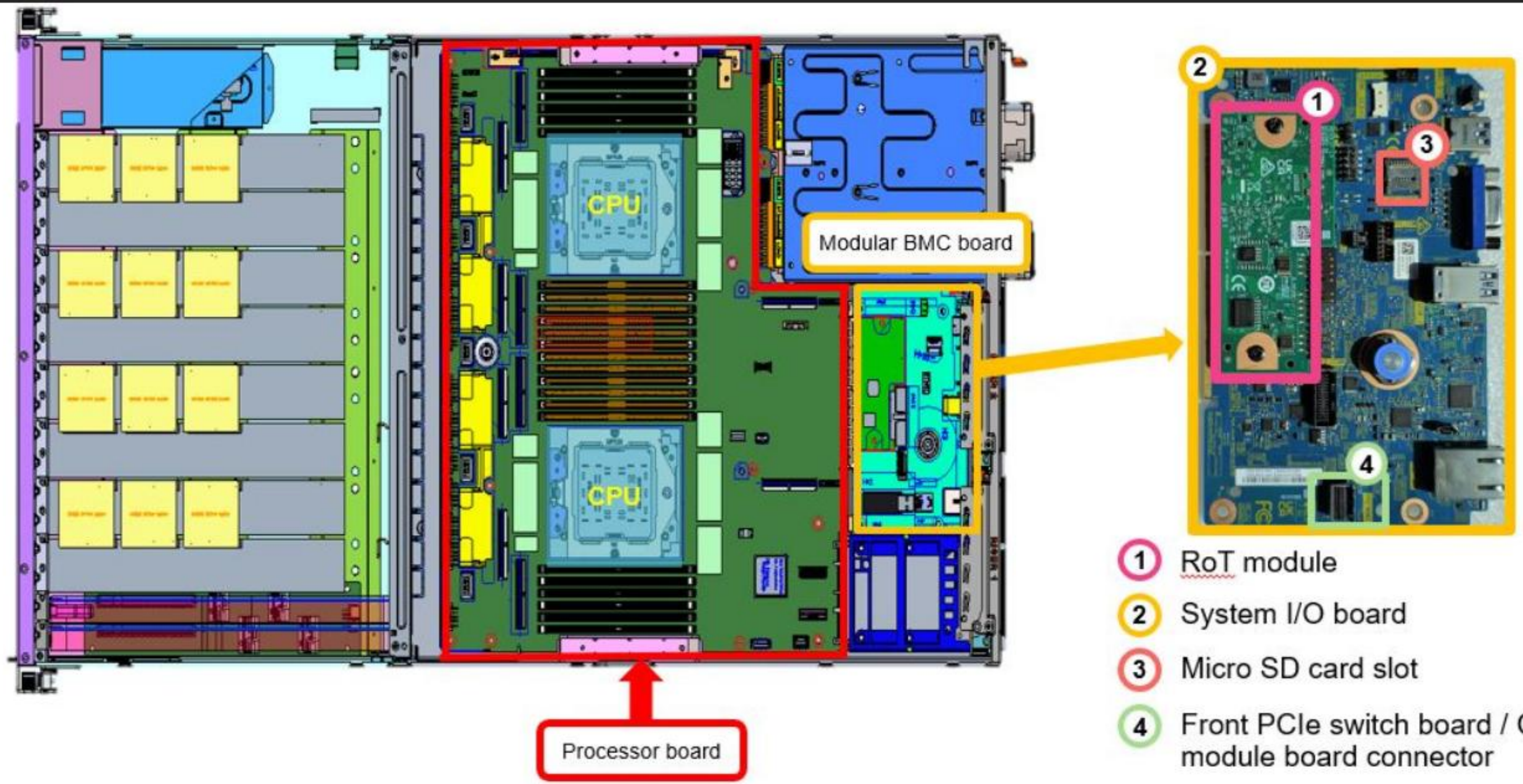
System board assembly

The SR675 V3 system board has three components:

- Processor board
  - A board containing CPU sockets, PCIe slots, memory slots, and other server component connectors
- System I/O board
  - A board containing the system BMC (XCC2) management port, USB ports, and a VGA connector
  - A Micro SD card slot to extend XCC2 storage space for the backup of firmware and for remote console virtual media
  - A signal connector to the front PCIe switch board or CX-7 module board (for SXM GPU configuration)
- Firmware and Root of Trust security module (RoT module)
  - A mezzanine card containing the Trusted Platform Module (TPM), UEFI firmware, XCC2 firmware, and a silicon Root of Trust

Click HERE to see the processor board, system I/O board, and RoT module locations.

Processor board, system I/O board, and RoT module



1. RoT module
2. System I/O board
3. Micro SD card slot
4. Front PCIe switch board / CX-7 module board connector

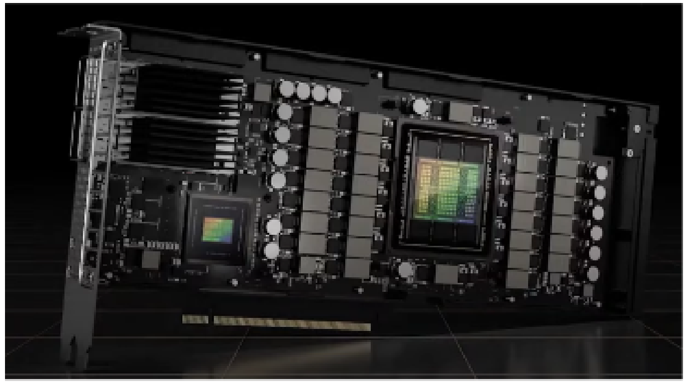
NIVIDA GPUs

The SR675 V3 supports the following types of GPUs:

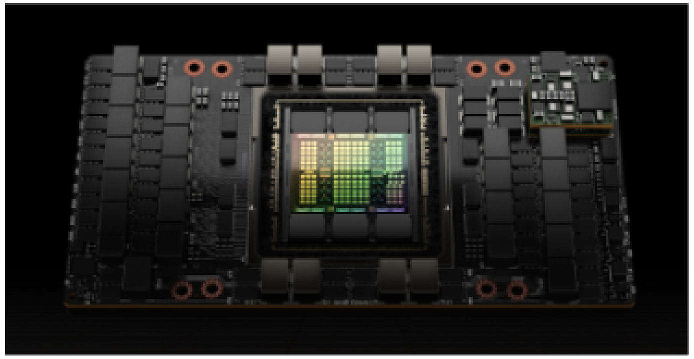
- PCIe GPU adapter
- PCIe GPU adapter with CNX technology
  - PCIe adapter with integrated NVIDIA CX-7 smart NIC
- SXM5 GPUs
  - Four NVIDIA HGX H100 or H200 GPUs installed on a GPU board – requires an LACM for cooling

For more information about the NVIDIA H100 GPU, refer to the following NVIDIA websites:

- https://www.nvidia.com/en-us/data-center/h100/
- https://www.nvidia.com/en-us/data-center/h200/



An NVIDIA GPU with CNX technology



An NVIDIA H100 GPU

PCIe switch board and CX-7 board for SXM GPUs

The SR675 V3 SXM GPU models support the following components for internal SXM GPU to system board assembly PCIe lane signal connections:

- PCIe switch board
  - This board has two PCIe switches
- NVIDIA CX-7 module board
  - This board has four NVIDIA Connect-X 7 (CX-7) modules and its carrier board

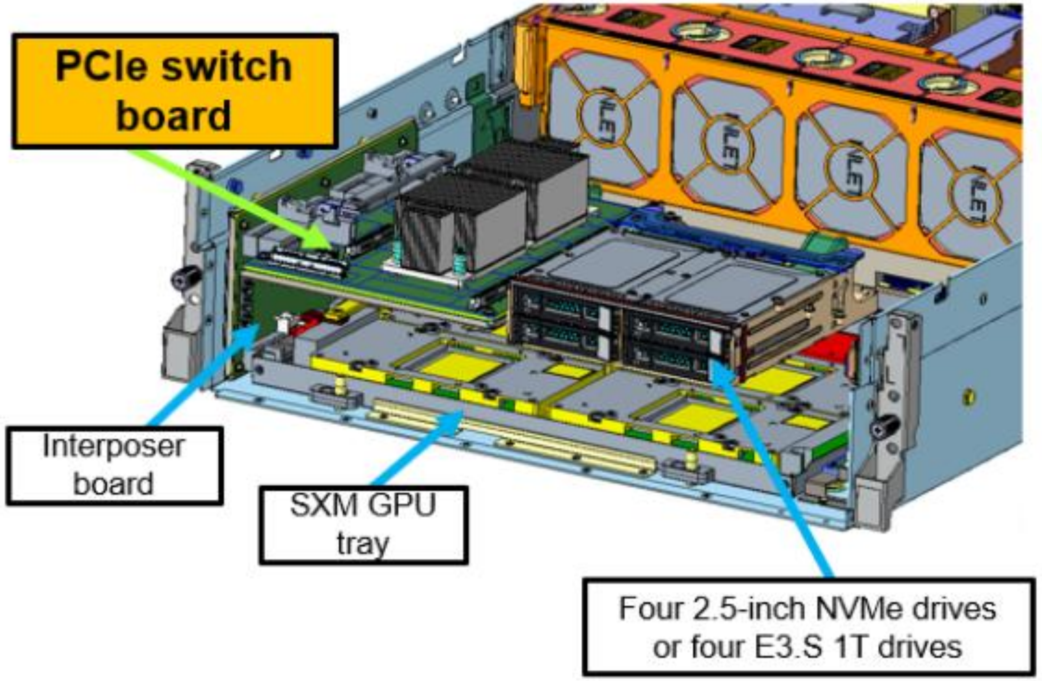
For SXM GPUs to work in the SR675 V3, one of these boards must be installed.

Click HERE to see the installation locations of the PCIe switch board and the CX-7 board in the front of an SR675 V3 SXM GPU model chassis.

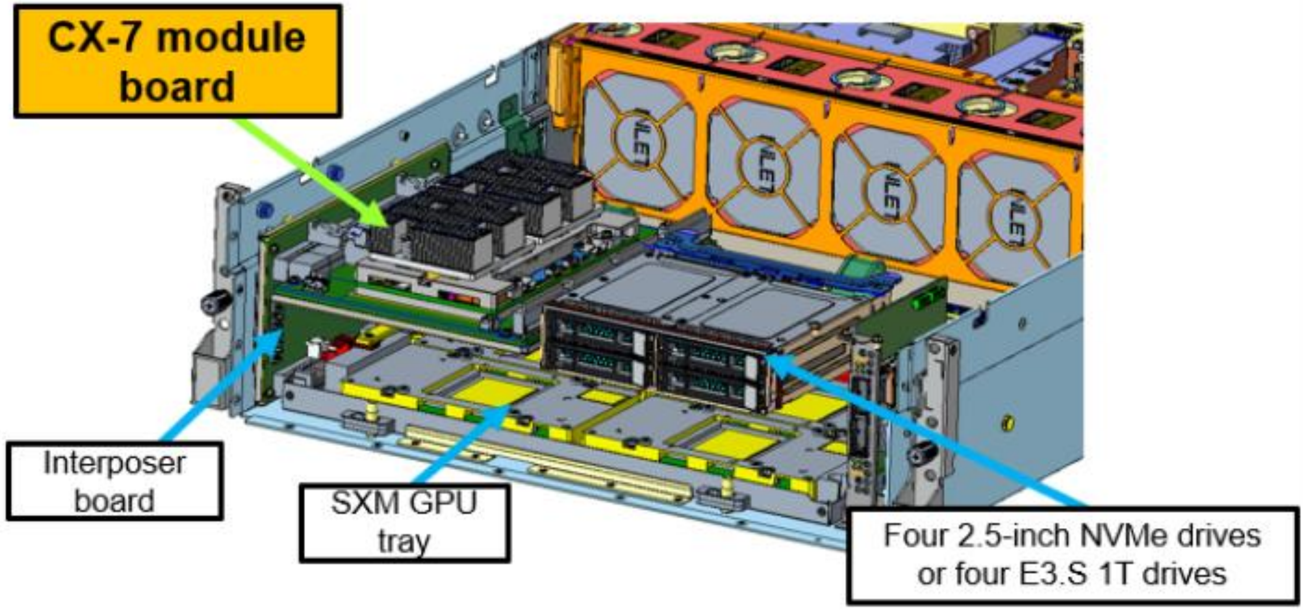
Click HERE to see schematic diagrams of the PCIe switch board and the CX-7 board.

SR675 V3 system components

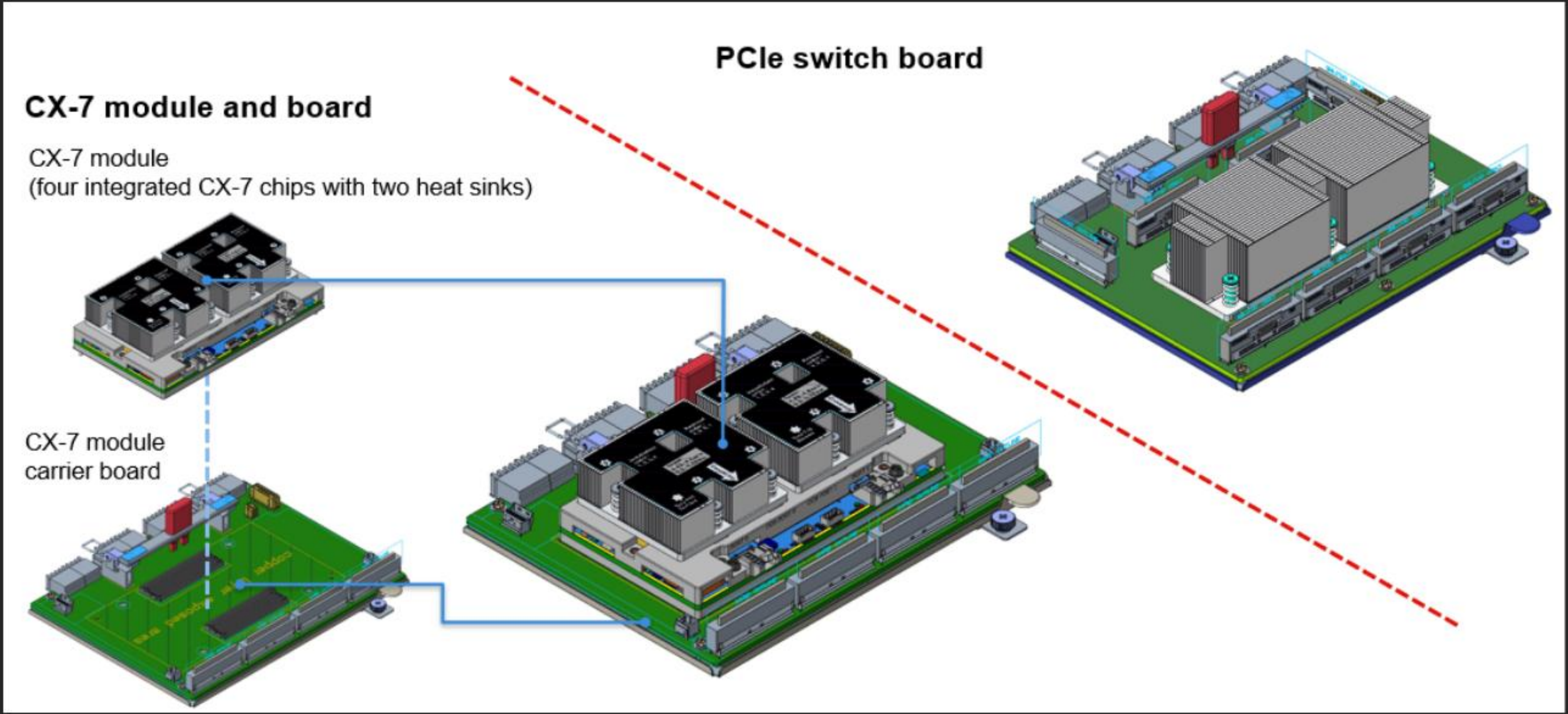
SXM GPU model with a PCIe switch board



SXM GPU model with a CX-7 module board



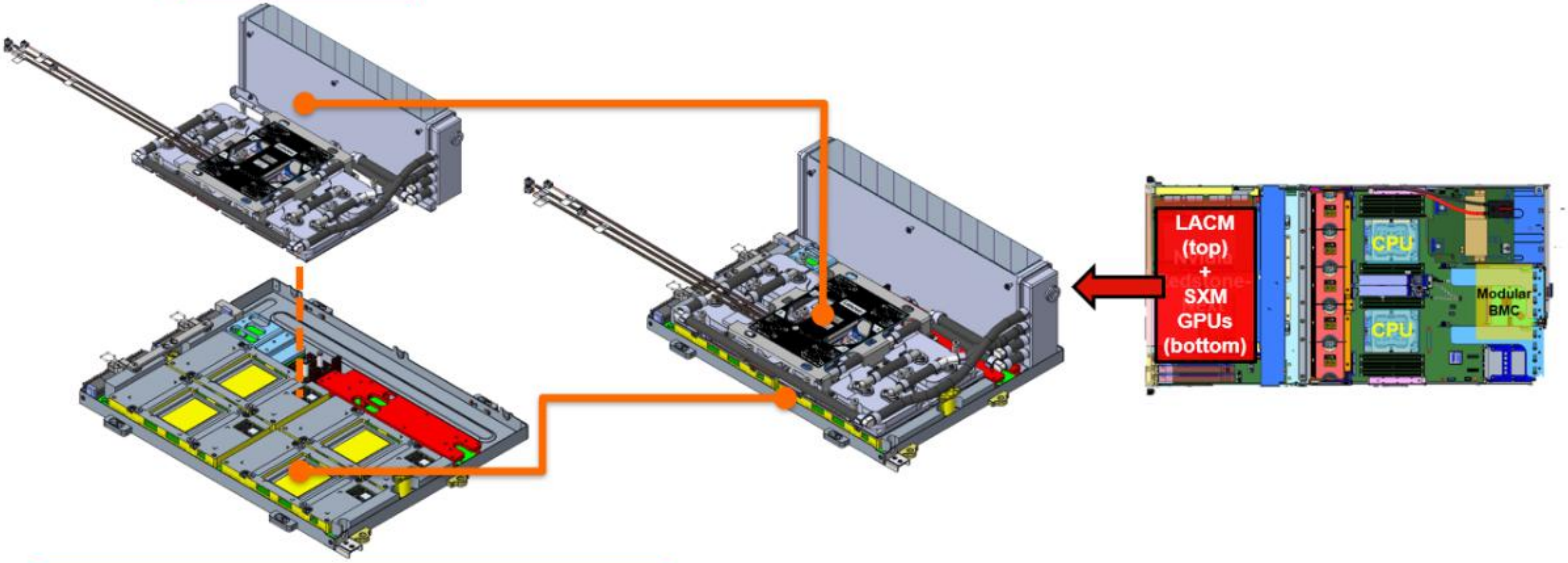
SR675 V3 system components



SXM GPU models with an LACM

The SR675 V3 supports the LACM for the cooling of NVIDIA SXM HGX H100 GPUs.

LACM

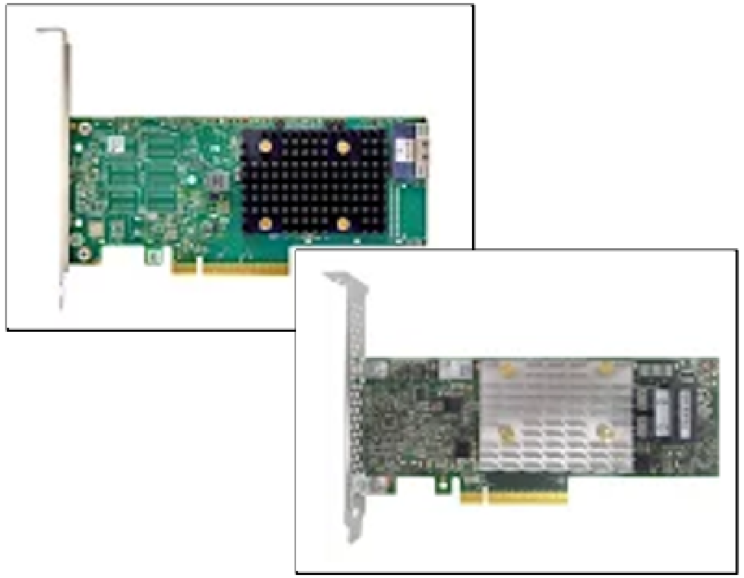


SXM GPU tray (four NVIDIA HGX H100 GPUs)

Storage controllers

The SR675 V3 supports the following storage controllers:

- ThinkSystem RAID 9350-8i 2 GB flash PCIe 12 Gb adapter
- ThinkSystem 440-16e SAS/SATA PCIe 4.0 12 Gb HBA
- ThinkSystem RAID 540-8i PCIe 4.0 12 Gb adapter
- ThinkSystem 440-8i SAS/SATA PCIe 4.0 12 Gb HBA
- ThinkSystem 440-8e SAS/SATA PCIe 4.0 12 Gb HBA
- ThinkSystem RAID 940-8e 4 GB flash PCIe 4.0 12 Gb adapter
- ThinkSystem RAID 5350-8i PCIe 12 Gb adapter
- ThinkSystem 4350-8i SAS/SATA 12 Gb HBA
- ThinkSystem RAID 940-8i 4 GB flash PCIe 4.0 12 Gb Adapter



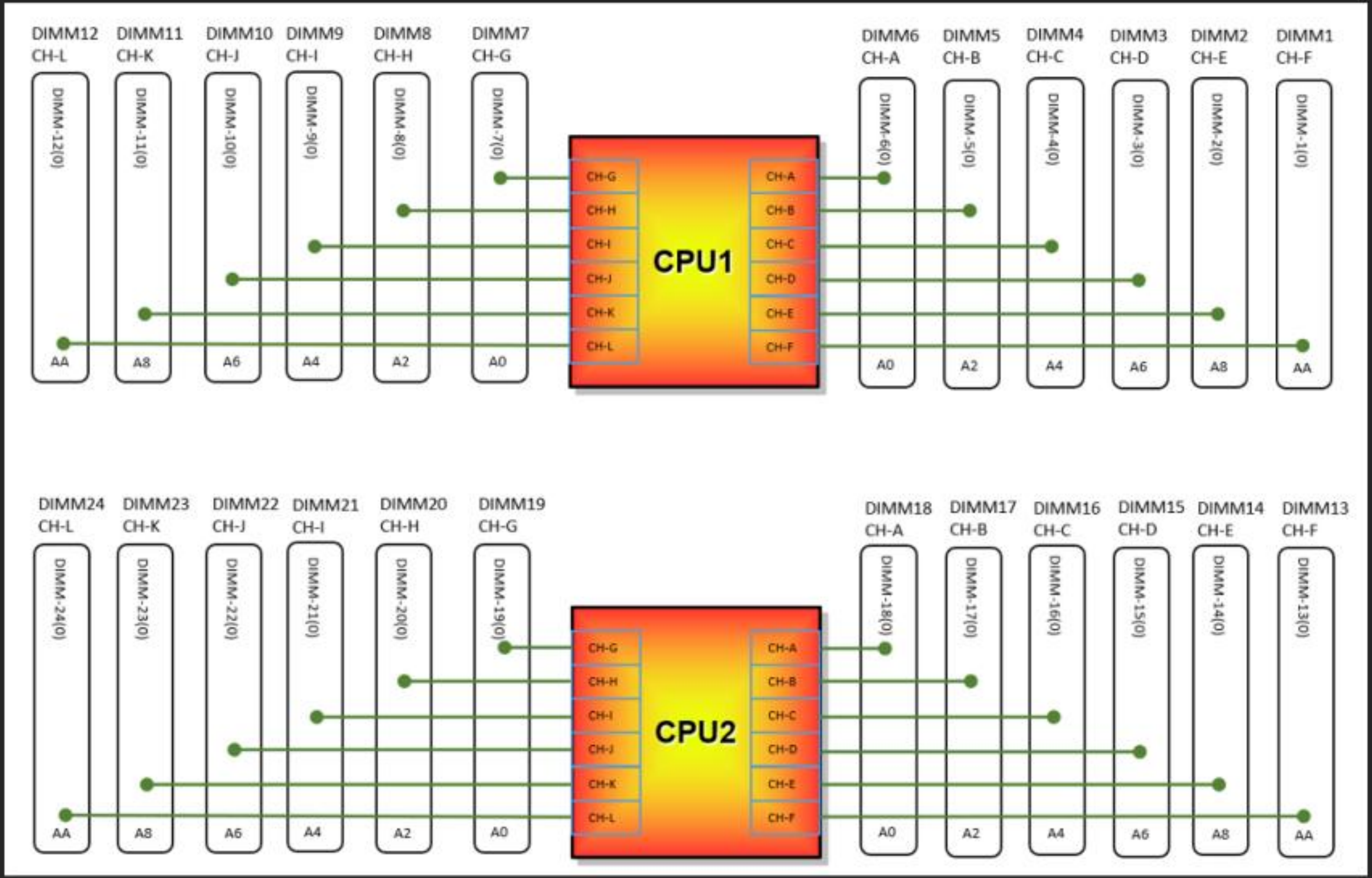
For the latest list of storage controllers supported by the SR675 V3, refer to the SR675 V3 Product Guide on Lenovo Press.

DIMMs

The SR675 V3 supports up to 24 DIMMs:

- One DIMM per channel
- Support for DDR5 RDIMMs up to 6000 MHz
- Support for RDIMMs (1Rx8, 2Rx4, and 2Rx8)
- Support for 3DS RDIMMs (2S2Rx4)
- Support for mixing of memory speeds
  - The system will operate at the lowest DIMM speed
- Support for a mixing of DIMM vendors
- DIMMs for each memory channel and CPU must have the same memory capacity and rank
- DIMMs must be installed in a specific order based on the system configuration
  For more information, refer to the Memory module installation rules and order section of the SR675 V3 user guide on the Lenovo Docs website
- Click HERE to see the SR675 V3 DIMMs block diagram

SR675 V3 DIMMs block diagram



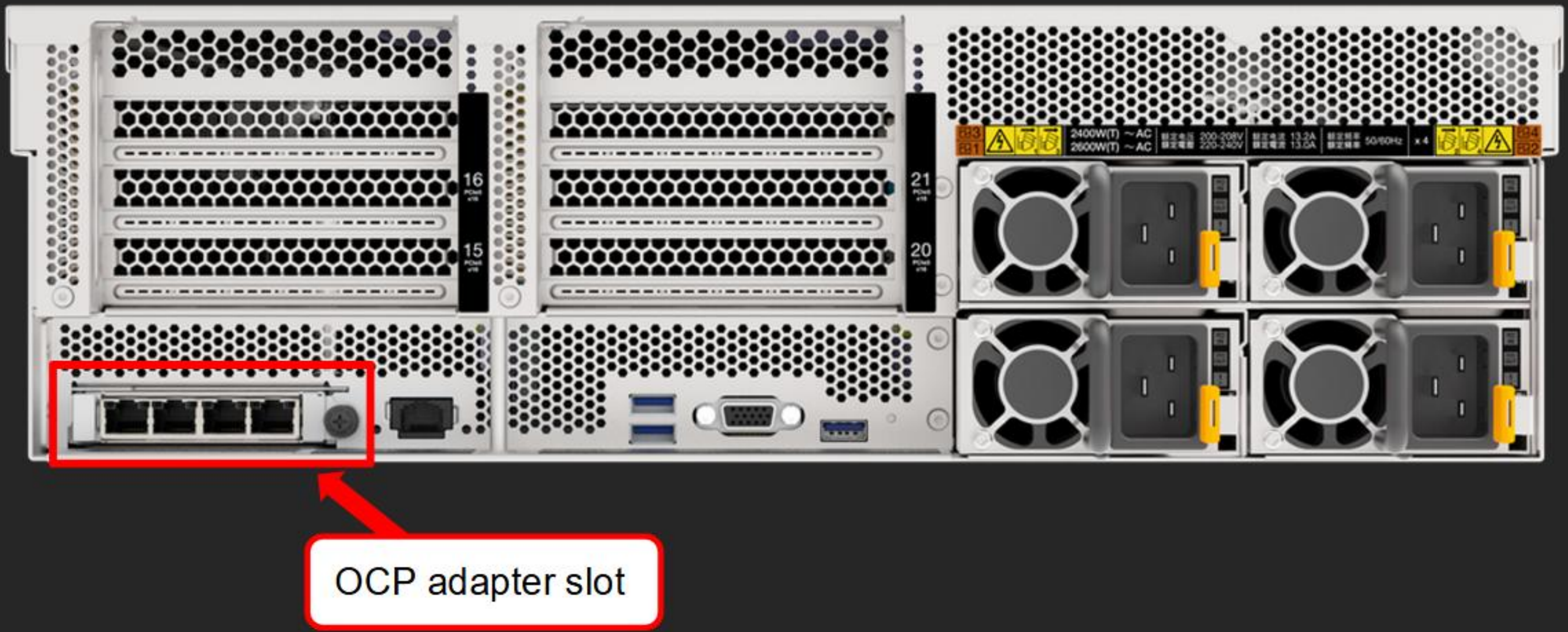
Network adapters

The SR675 V3 supports the following types of network adapter:

- OCP adapters
  - PCIe 4.0
  - Installed in the rear OCP adapter slot
  - Click HERE to see OCP adapter slot locations
- Standard form-factor PCIe network adapters are supported in PCIe slot 1, 2, 15, 16, 20, and 21
  - Slots 1 and 2: FHFL adapters in 8-DW and 4-DW GPU configurations
  - Slots 1 and 2: FHHL in SXM5 GPU configurations
  - Slots 15, 16, 20, and 21 support FHHL adapters for all SR675 V3 configurations
  - For PCIe slot locations, refer to the System board assembly connectors section of the SR675 V3 User Guide

For the latest list of network adapters supported by the SR675 V3, refer to the network adapter section of the SR675 V3 product guide

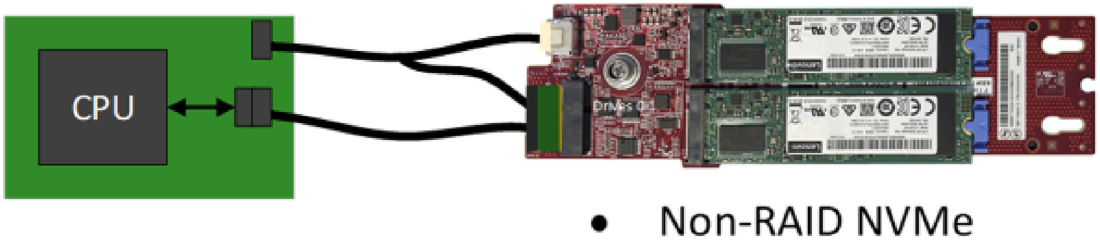
SR675 V3 system components



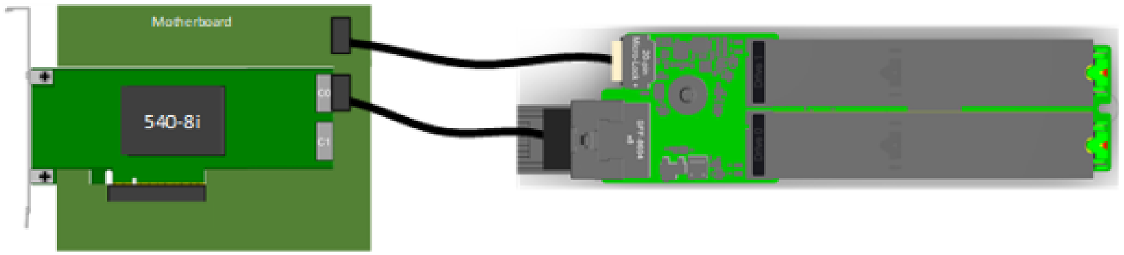
M.2 adapters

The SR675 V3 supports non-RAID NVMe and RAID NVMe M.2 boot solutions:

- The NVMe M.2 adapter with non-RAID configuration is supported with the Non-RAID SATA/NVMe boot kit attached to the CPU
- The NVMe M.2 adapter with RAID configuration is supported with the Non-RAID SATA/x4 NVMe boot kit attached to a ThinkSystem 540-8i RAID controller



Non-RAID NVMe M.2



RAID NVMe M.2